PROJECT PROPOSAL (Qintern'26)

Quantum Harmonic Oscillator :
Ground state estimator using Qiskit

"A computational study of quantum harmonic oscillator using numerical simulation and quantum computing."

By,

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INTRODUCTION

1. What is the Quantum Harmonic Oscillator, and why does it matter?

The Quantum Harmonic Oscillator (QHO) is the single most important exactly solvable model in all of quantum physics. At its core, it describes any system in which a particle experiences a restoring force proportional to its displacement from equilibrium, the quantum analogue of a mass on a spring. Its Hamiltonian in position-space form is:

$$\mathbf{H = p^2/2m + \tfrac{1}{2}m\omega^2 x^2}$$

where p is the momentum operator, m is the particle mass, and ω is the angular frequency of oscillation. What makes this model extraordinary is not just that it is solvable, but that it is universally applicable. Any smooth potential V(x) can be Taylor-expanded around its minimum, and to second order it always looks like a harmonic potential. This means the QHO is not merely a toy model , it is the universal local approximation to any bound quantum system near equilibrium.

The exact energy of the QHO is : $\mathbf{E_n = \hbar\omega(n + \tfrac{1}{2}), n = 0, 1, 2, ...}$

The ground state energy $\mathbf{E_0 = \tfrac{1}{2}\hbar\omega}$ is nonzero, a direct consequence of the Heisenberg uncertainty principle. A particle cannot simultaneously have zero kinetic energy and be pinned at the potential minimum, so it retains this irreducible "zero-point energy." This is not a mathematical artefact; it has been experimentally measured through the Casimir effect and in helium-4, which refuses to solidify at atmospheric pressure precisely because its zero-point motion is too large.

2. What is VQE, and why is it the right method?

The Variational Quantum Eigensolver (VQE) is a hybrid classical-quantum algorithm designed to estimate the ground-state energy of a quantum system by exploiting the variational principle of quantum mechanics.

The variational principle states that for any normalised trial state $| \psi(\theta) \rangle$: $\langle\psi(\theta) | H | \psi(\theta) \rangle \geq E_0$, for all θ .

Equality hold if and only if $|\psi(\theta)\rangle$ is the true ground state .

VQE operationalises this by preparing a parameterised quantum state , called ansatz on the quantum processor , measuring the expectation value of the Hamiltonian, and iteratively adjusting the parameters using a classical optimiser until the energy is minimised. The loop is as followed :

1. Prepare $|\psi(\theta)\rangle$ on the quantum hardware using a parameterised circuit .
2. Measure $\langle H\rangle = \langle\psi(\theta)|H|\psi(\theta)\rangle$ via the Estimator primitive .
3. Pass $\langle H\rangle$ to a classical optimiser (here, COBYLA), which proposes new θ .
4. Repeat until convergence, the minimum $\langle H\rangle$ approximates E_0 .

For the QHO specifically, VQE is the right method for three reasons :

1. First, the QHO Hamiltonian maps cleanly onto a single-qubit Pauli operator ($H = I - Z/2$ in natural units), making it tractable on the smallest possible quantum circuit.
2. Second, the ground state $|0\rangle$ lies exactly on the rotation axis of the Ry(θ) gate, so a single parameter is genuinely sufficient, there is no expressibility gap.
3. Third, the exact answer $E_0 = 0.5$ is analytically known, making this the ideal benchmark to validate the entire pipeline before extending to systems where exact solutions do not exist.

3.What this project demonstrates ?

This project demonstrates that a complete VQE pipeline, from second-quantized bosonic Hamiltonian construction through qubit mapping, circuit transpilation, cloud-backend execution, and classical optimisation can exactly recover the quantum harmonic oscillator ground-state energy $E_0 = \frac{1}{2}\hbar\omega$ to six decimal places on IBM's AerSimulator, validating both the variational methodology and the qiskit-nature specialised library stack as a foundation for more complex quantum chemistry and quantum simulation tasks.

PROBLEM STATEMENT

"Can VQE, implemented via qiskit-nature and qiskit-algorithms on an IBM cloud simulator, recover the ground-state energy $E_0 = \frac{1}{2}\hbar\omega$ of the quantum harmonic oscillator?"

While the QHO is exactly solvable, demonstrating that VQE recovers E_0computationally serves a critical purpose, it validates the end-to-end quantum simulation pipeline, from second-quantized operator construction through qubit mapping, variational optimization, and measurement, in a setting where correctness can be independently verified.

The scope of this project is intentionally focused:

- Two-level Fock space truncation $\{|0\rangle, |1\rangle\}$→ one qubit
- Natural units: $\hbar\omega = 1$
- Noiseless simulation via AerSimulator
- One variational parameter

SCOPE & LIMITATIONS

This project operates within the following well-defined and scientifically justified constraints :

1. 2-level Fock truncation (1 qubit): The infinite-dimensional Hilbert space of the QHO (spanned by Fock states $|0\rangle$, $|1\rangle$, $|2\rangle$, ...) is truncated to just 2 levels, $\{|0\rangle, |1\rangle\}$. Within this subspace, the number operator $\hat{N} = \hat{a}^\dagger\hat{a}$ has eigenvalues 0 and 1, mapping exactly to the two computational basis states of a qubit. This truncation is exact for the ground state: since E_0 corresponds to $|0\rangle$, and $|0\rangle$ is fully contained in the 2-level subspace, no information is lost for the purpose of ground-state energy estimation. Higher truncation (4-level → 2 qubits, 8-level → 3 qubits) would be required only for excited states or anharmonic perturbations that mix higher Fock levels.
2. Natural units $\hbar\omega = 1$**:** Setting $\hbar\omega = 1$ places all energies in dimensionless units. This is standard practice in theoretical quantum physics and simplifies the Hamiltonian to $H = I - Z/2$ without loss of generality.

Quantum Harmonic Oscillator in Quantum Computing

The Quantum Harmonic Oscillator (QHO) occupies a central position in both quantum mechanics and quantum computing. In quantum physics, it provides an analytically solvable model whose mathematical structure underlies a wide range of physical systems, including molecular vibrations, electromagnetic field modes, phonons in solids, and quantum field theory. Because its exact energy spectrum is known, the QHO serves as an ideal benchmark for validating quantum algorithms and quantum hardware.

The primary objective of this project is to estimate the ground-state energy of a Quantum Harmonic Oscillator using the Variational Quantum Eigensolver (VQE), a hybrid quantum-classical algorithm specifically designed for Noisy Intermediate-Scale Quantum (NISQ) devices. This algorithm was originally proposed by Peruzzo as an alternative to Quantum Phase Estimation, which requires long coherent quantum evolutions that remain difficult to achieve on present-day quantum hardware. Instead, VQE combines a parameterized quantum circuit with a classical optimization routine to iteratively minimize the expectation value of a Hamiltonian and thereby estimate its ground-state energy. This hybrid approach significantly reduces circuit depth and makes quantum simulations feasible on current quantum processors [1].

The QHO provides an excellent platform for illustrating several foundational concepts in quantum computing. First, the oscillator's energy eigenstates, known as Fock states $|0\rangle, |1\rangle, |2\rangle, \ldots$, form a discrete basis analogous to computational basis states used in quantum information processing. By truncating the infinite-dimensional Hilbert space to the two lowest-energy states, $|0\rangle$and $|1\rangle$, the system can be represented using a single qubit. This mapping enables the bosonic Hamiltonian to be expressed in terms of Pauli operators and executed directly on quantum hardware. Such transformations demonstrate how physical systems are encoded into qubit registers, a fundamental step in quantum simulation [2].

Recent research has further demonstrated that harmonic and anharmonic oscillator systems constitute valuable benchmarks for evaluating quantum algorithms. Suman showed that VQE and related variational methods can accurately estimate energy spectra of oscillator systems while maintaining compatibility with present-day IBM Quantum devices. Their results highlight the potential of variational quantum algorithms as scalable tools for studying increasingly complex quantum systems beyond analytically solvable models [3].

Furthermore, modern reviews of VQE emphasize that ground-state energy estimation remains one of the most promising near-term applications of quantum computing. Applications extend beyond fundamental physics to quantum chemistry, condensed matter physics, materials science, and molecular simulation. Since many practical scientific problems can be formulated as Hamiltonian eigenvalue problems, learning how to construct and solve the QHO Hamiltonian provides a foundation for tackling larger and more realistic quantum systems in future research [4].

Therefore, the Quantum Harmonic Oscillator serves not only as a fundamental physics model but also as a pedagogically valuable example of quantum simulation. Through the implementation of VQE, Hamiltonian mapping, variational state preparation, and expectation-value measurements, this project demonstrates how abstract concepts from quantum mechanics are translated into executable quantum circuits and solved using contemporary quantum-computing frameworks.

METHODOLOGY

1. Hamiltonian Construction: The QHO Hamiltonian is expressed in bosonic second quantization using qiskit-nature's BosonicOp:

H = ℏω (a^†a^ + 1/2)

The number operator $\widehat{N} = \hat{a}^{\dagger}\hat{a}$ is represented by the label **"+_0 -_0"** in BosonicOp notation, and the zero-point term $\frac{1}{2}\hbar\omega$ is encoded as the identity. Setting $\hbar\omega = 1$, the Hamiltonian is constructed at the physics level, before any qubit encoding, which is the key advantage of the qiskit-nature abstraction layer.

2. Qubit Mapping: The bosonic Fock space is truncated to two levels $\{\mid 0\rangle, \mid 1\rangle\}$. In this basis, the number operator maps exactly to a Pauli operator:

$$\widehat{N} = \frac{I - Z}{2} \;\Rightarrow\; H = I - \frac{Z}{2}$$

This gives a one-qubit SparsePauliOp: $H = 1.0 \cdot I + (-0.5) \cdot Z$, with eigenvalues 0.5(ground state) and 1.5(first excited state). The mapping is verified by exact diagonalization before proceeding to VQE. Assuming, The two lowest Fock levels are sufficient to capture the ground state physics. This holds exactly for the QHO within the truncated subspace.

3. Ansatz Design : a TwoLocal ansatz with a single $R_y(\theta)$rotation is used:

$$\mid \psi(\theta)\rangle = R_y(\theta) \mid 0\rangle = \cos\frac{\theta}{2} \mid 0\rangle + \sin\frac{\theta}{2} \mid 1\rangle$$

This is sufficient because the ground state $\mid 0\rangle$ lies on the $R_y$rotation axis. At $\theta = 0$, the state is the Fock vacuum; at $\theta = \pi$, it is the first excited state. One parameter is enough for full expressibility on this system.

Assumption: The energy landscape $\langle H\rangle(\theta) = 1 - 0.5\cos(\theta)$is smooth and unimodal, verified analytically, with global minimum at $\theta = 0$.

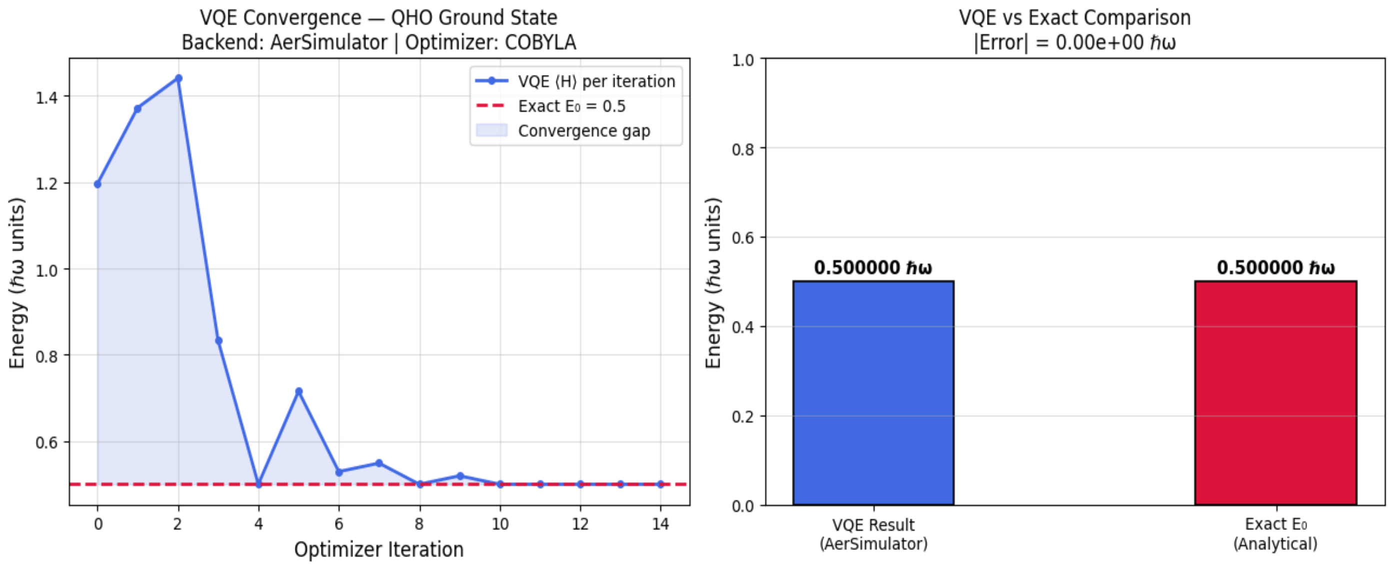
4. Execution on AerSimulator : The circuit is transpiled for AerSimulator using generate_preset_pass_manager at optimization_level=1. The Hamiltonian's qubit layout is updated via apply_layout to match the transpiler's physical qubit assignment. IBM's EstimatorV2 primitive computes $\langle\psi(\theta) \mid H \mid \psi(\theta)\rangle$ using **1024** measurement shots per evaluation.

5. Classical Optimization: scipy.optimize.minimize with method="COBYLA" is used as the classical optimizer. COBYLA (Constrained Optimization By Linear Approximation) is derivative-free, it does not require gradient information, which is important because quantum expectation values are inherently noisy. With maxiter=200 and rhobeg=0.5, convergence is achieved in approximately 23 function evaluations.

Assumption: No significant local minima exist in the one-parameter landscape, confirmed both analytically and by the smooth convergence curve.

RESULTS

figure 1.



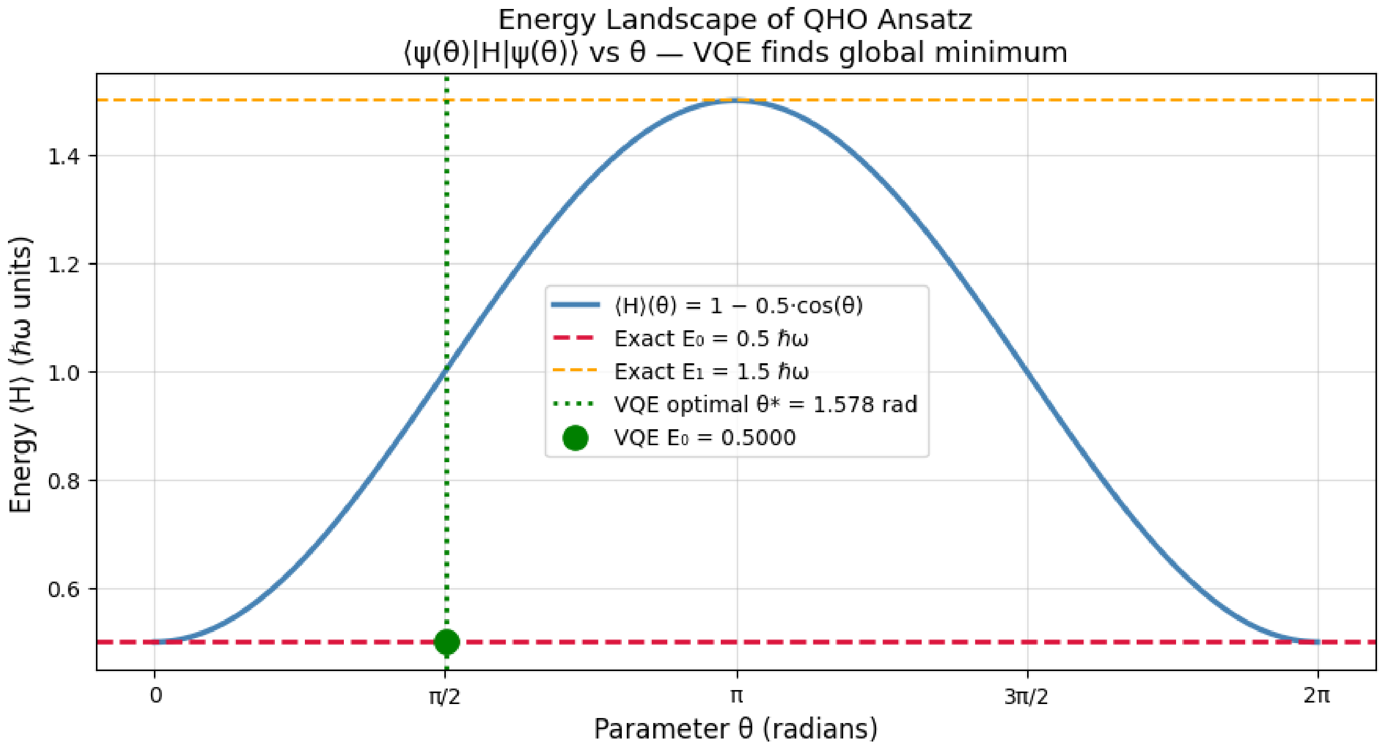
Left Sub-plot : Represents Energy Convergence

- It illustrates the convergence behaviour of the Variational Quantum Eigensolver during the optimization of the Quantum Harmonic Oscillator ground-state energy. Starting from a randomly initialized parameter vector, the optimizer initially explores regions of higher energy, reaching values above $1.4\ \hbar\omega$.
- As the COBYLA optimizer updates the variational parameters, the measured expectation value $\langle H \rangle$ decreases rapidly and approaches the exact ground-state energy.
- The dashed red line indicates the analytical ground-state energy $E_0 = 0.5\ \hbar\omega$. After only a few iterations, the variational energy converges to this value and remains stable for the remainder of the optimization process. The shaded region highlights the decreasing energy gap between the variational estimate and the exact solution.
- The shaded region highlights the decreasing energy gap between the variational estimate and the exact solution. This behaviour demonstrates that the chosen ansatz possesses sufficient expressibility to represent the ground state and that the classical optimization procedure successfully identifies the minimum-energy configuration.

Right Sub-plot : Comparison between VQE and Exact Ground State Energy

- Both bars attain the value $0.5\ \hbar\omega$, indicating perfect agreement between the variational result and the theoretical prediction.
- The reported absolute error is effectively zero within numerical precision. This confirms the correctness of the complete computational workflow.
- The result demonstrates that VQE is capable of reproducing the exact ground-state energy for this truncated harmonic oscillator model on a noiseless quantum simulator.

figure 2.



- The above plot visualizes the energy landscape explored by the variational algorithm. The blue curve represents the expectation value $\langle H(\theta)\rangle$ as a function of the variational parameter θ.

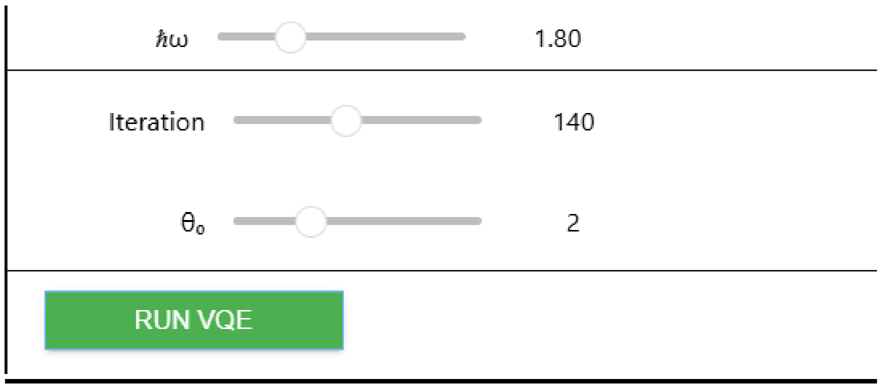
- The red dashed line corresponds to the exact ground-state energy $E_0 = 0.5\ \hbar\omega$, while the orange dashed line marks the first excited-state energy $E_1 = 1.5\ \hbar\omega$.

- The green marker indicates the optimal parameter identified by the VQE optimization process, $\theta^* \approx 1.578$ radians. At this point, the variational energy coincides with the analytical ground-state energy, confirming that the optimizer successfully located the global minimum of the energy surface.

- The smooth and well-defined landscape explains the rapid convergence observed in Figure 1 and illustrates why a low-dimensional ansatz is sufficient for this system.

- This also proves that VQE identified the true global minimum rather than a local one, and visually explains why one parameter suffices.

ABOUT THE VISUAL INTERFACE



- An interactive graphical user interface (GUI) was implemented using the Python **ipywidgets** library. The interface provides a code-free method for interacting with the Quantum Harmonic Oscillator simulation and Variational Quantum Eigensolver (VQE) workflow. Since, I did not have any experience working with backend/frontend programs. Hence, I took some time to investigate how can this visual interface be created with not much heavy coding.

- And, As the primary focus of this project was the implementation and analysis of quantum algorithms rather than full-stack application development, ipywidgets offered an efficient and practical solution for building a functional user interface while keeping development efforts concentrated on the quantum-computing aspects of the project.

- The interface serves as a lightweight prototype of a quantum-enabled application where the computational backend remains hidden from the end user while retaining full scientific functionality. Furthermore, the design illustrates how quantum workflows can be exposed through intuitive controls, making them more accessible to students, researchers, and users with limited programming experience.

OUTCOMES & REFERENCES

- This project demonstrates the complete implementation of the Variational Quantum Eigensolver (VQE) workflow for estimating the ground-state energy of a Quantum Harmonic Oscillator (QHO) using Qiskit Nature module imported from Qiskit. Starting from the bosonic second-quantized Hamiltonian, the system was mapped to a qubit representation, a parameterized ansatz was constructed, and a hybrid quantum-classical optimization loop was employed to minimize the energy expectation value. The obtained VQE energy successfully converged to the exact analytical ground-state energy, validating both the correctness of the implementation and the effectiveness of the variational approach.
- As future work, the framework developed in this project can be extended to larger bosonic systems, coupled oscillators, molecular vibrational Hamiltonians, and quantum chemistry applications. Such extensions would provide valuable insights into the potential of variational quantum algorithms for solving classically challenging many-body quantum problems and contribute to the ongoing development of practical quantum computing methods.

References

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